Uncertainty-Aware Skin Cancer Classification using Two-Stage CNN, MC Dropout and Explainable AI

Rohini (202311019), Jasmeen Kaur (202311037), Hasini Reddy (202311040)
Under the guidance of Dr. Jignesh Patel
Indian Institute of Information Technology Vadodara (IIITV-ICD)

***Abstract*—Deep learning models have made great strides in classifying medical images, especially for detecting skin cancer. However, these models often struggle with reliability and interpretability, which are essential in healthcare. In this study, we create a framework for uncertainty-aware classification based on AUQantO, using a two-stage Convolutional Neural Network (CNN). We use Monte Carlo (MC) Dropout to estimate predictive uncertainty and apply COBYLA optimization to find a suitable uncertainty threshold for selective prediction. We also add explainable AI techniques like Integrated Gradients and spatial uncertainty maps to help visualize model decisions. The experimental results show that rejecting uncertain predictions boosts accuracy from 80.3% to 82.45% and offers valuable insights into how the model operates.**

I. Introduction

Skin cancers are one of the most prevalent kinds of cancer worldwide. Beneficial for treatment is the early identification of skin cancers. Recently, researchers demonstrated that deep-learning algorithms provide accurate classification of skin lesions as either benign or malignant. However, many of these algorithms are typically viewed as "black boxes," thus not providing any indication of how confident they are in their final diagnosis; therefore, limiting their reliability in the clinic.

In terms of medical applications, accurate diagnosis is not enough; you also need to determine if symptoms are reasonably predictable. Hence, there has been a demand for uncertainty aware learning methods.

Recent developments in uncertainty quantification have enabled models to produce estimates of their confidence using various techniques (i.e. Monte Carlo Dropout) with many advantages. In particular, they allow users to:

- Make predictions with greater confidence
- Perform Bayesian Inference with Deep Neural Networks
- Identify unreliable predictions for expert assessments
- Improve public trust in safety-critical environments like healthcare

Besides reliability, explainability will be key for acceptance into the clinical environment. In particular, Explainable AI will aid in interpreting a model's decision by providing methods to:

- Show important parts of an image (i.e. Use Integrated Gradients)
- Identify problematic areas (i.e. Produce spatial uncertainty maps)
- Facilitate decision-making through transparency

In this work, we propose a unified framework that integrates:

- A Two-stage CNN to extract features from image patches
- Monte Carlo Dropout for uncertainty quantification
- COBYLA optimization for selecting feature thresholds
- Explainable AI techniques for encouraging trust in the model

The added functionality provided by the new system creates a more reliable model by allowing it to refrain from making potentially unreliable predictions, thus increasing the potential for real world medical use by increasing public trust and confidence in the system.

II. Related Work

A. Deep Learning for Medical Image Classification

There are multiple deep learning techniques for predicting output images using medical imaging methods such as Convolutional Neural Networks (CNN). Commonly used methods include:

- **Patch-based CNNs:** Two-Stage CNN and DSF-CNN both methods identify local features by breaking down an image into segments (patches), they use a two-phase deep learning approach.
- **Ensemble Models:** which rely on combining multiple deep learning models to increase model robustness.
- **Hybrid Architectures:** Combine CNNs with models such as LSTMs.

B. Uncertainty Quantification

Additionally, uncertainty estimation allows medical professionals to determine whether a prediction is reliable. Common methods for estimating uncertainty include the following:

- **MC Dropout:** provides a Bayesian approximation to Bayesian inference through the use of multiple forward runs of the model.
- **Entropy-based Methods:** measures the level of ambiguity in a particular prediction.
- **Variance-based Measures:** Variance of predictions across multiple forward passes through the model.

C. Existing Uncertainty-Aware Models

There are currently numerous works that incorporate uncertainty into their CNN architecture. Examples include:

- **Bayesian CNNs and DenseNet:** both of which incorporate dropout as a method to estimate uncertainty.
- **ARA-CNN:** which incorporates uncertainty models into residual learning.

- **Instability Maps:** which highlight areas of ambiguity in an image.

Despite these capabilities, most existing uncertainty-aware models are geared towards estimating uncertainty rather than utilizing it as part of the prediction process.

D. Limitations of Existing Work

Existing studies have the following limitations:

- No actionable information (can't reject uncertain samples)
- Requires certain architectures to work
- Can't apply the results generally

E. AUQantO Framework

AUQantO framework resolves these limitations by:

- Has a model-agnostic wrapper that is not dependent on models or specific architecture
- Estimates uncertainty by using MC Dropout and entropy
- Optimizes (COBYLA) and learns a rejection threshold
- Facilitates automatic exclusion of uncertain samples

By focusing on high-confidence predictions, we will be able to improve reliability.

F. Explainable AI

Explainable AI methods are used to make machine learning more interpretable by:

- Using Integrated Gradients to indicate the most significant image areas impacting the predicted outcome
- Using Spatial Uncertainty Maps to indicate the image areas with the most uncertainty about the predicted outcome

Combining uncertainty with interpretability increases the overall transparency of medical-based AI systems.

III. Methodology

A. Two-Stage CNN Architecture

The model uses a two-step approach to classify images, focusing first on small details and then on the bigger picture.

Patch-level CNN:

- Each image is split into smaller, overlapping patches of the same size.
- These patches are processed one at a time using a convolutional neural network (CNN).
- The CNN includes convolutional layers with ReLU activation, along with dropout and max-pooling to improve learning and prevent overfitting.
- Finally, adaptive average pooling is applied to compress the feature maps into compact representations.

Image-level Classifier:

- The features extracted from all patches are reorganized so they correspond to their original image.
- To summarize the information, both mean pooling and max pooling are applied across all patches.
- These combined features are then passed through fully connected layers to produce the final classification.

This design helps the model capture fine-grained local details as well as overall image patterns.

B. Monte Carlo Dropout

To measure how confident the model is in its predictions, dropout is kept active during testing. This introduces randomness and allows the model to generate slightly different outputs each time.

- The model performs multiple forward passes (T times) for the same input.
- Each pass produces a slightly different prediction due to dropout.
- The final prediction is computed by averaging all outputs:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^{T} p(y|x, w_t) \tag{1}$$

- Uncertainty is estimated using entropy:

$$H(p) = -\sum p \log p \tag{2}$$

Higher entropy means the model is less certain about its prediction.

C. Threshold Optimization

To make predictions more reliable, a threshold λ is used to separate confident predictions from uncertain ones.

- Accepted samples: $unc < \lambda$
- Rejected samples: $unc \geq \lambda$

The aim is to keep accurate predictions while filtering out unreliable ones.

- The optimization objective is defined as:

$$F(\lambda) = H_{exc} + (1 - H_{inc}) \tag{3}$$

where:

- H_{inc} is the average loss for accepted samples
- H_{exc} is the loss for rejected samples
- The COBYLA optimization algorithm is used to find the best value of λ, while ensuring the rejection rate stays within reasonable limits.

This approach allows the model to automatically identify uncertain predictions and avoid making unreliable decisions, improving overall performance.

IV. Explainability

A. Integrated Gradients

Integrated Gradients (IG) helps show which parts of an image influence the model's decision the most by assigning importance to each pixel.

- It starts with a baseline image (usually a completely black image) as a reference point.
- The original image is then gradually built up from this baseline.
- At each step, gradients are calculated and averaged to estimate how much each pixel contributes to the final prediction.

The attribution is computed as:

$$IG = (x - x') \times \int_0^1 \frac{\partial F(x)}{\partial x} d\alpha \tag{4}$$

- x is the original input image and x' is the baseline.
- $F(x)$ represents the model's output for the target class.

The final result is a heatmap that highlights the regions of the image that had the strongest impact on the model's decision, making the prediction easier to understand.

B. Spatial Uncertainty

Alongside pixel-level explanations, spatial uncertainty highlights where the model is unsure about its predictions.

- The image is divided into smaller patches.
- For each patch, multiple predictions are generated using Monte Carlo Dropout.
- The variation between these predictions is used to measure uncertainty.

Uncertainty for each patch is calculated as:

$$\sigma^2 = Var(predictions) \tag{5}$$

- Higher variance means the model is less certain.
- Lower variance means the model is more confident.

These uncertainty values are mapped back onto the image as a heatmap, showing which regions the model finds ambiguous or difficult.

C. Combined Interpretation

Bringing these two methods together gives a clearer picture of how the model behaves:

- Integrated Gradients shows **where the model is looking.**
- Spatial uncertainty shows **where the model is unsure.**

Together, they provide a more complete and transparent explanation of the model's predictions, helping users better understand and trust its decisions.

V. Dataset and Implementation

The model is trained on a skin cancer dataset with two categories: benign and malignant. It consists of dermoscopic images of skin lesions, which are resized and preprocessed before being fed into the model.

- Image size: 224×224
- Patch size: 112×112 (with overlapping patches extracted from each image)
- Batch size: 2
- Learning rate: $1e^{-4}$
- MC runs: 50
- To improve the model's ability to generalize, data augmentation techniques such as image flipping and rotation are applied during training.
- Training is carried out in two stages. First, the patch-level CNN is trained to learn local features. Then, these features are used by the image-level classifier to make the final prediction.
- During inference, dropout remains active to enable Monte Carlo Dropout, which allows the model to estimate prediction uncertainty.

The entire framework is implemented using PyTorch.

VI. Results

A. Qualitative Results

To better understand how the model behaves, we examine examples of accepted and rejected predictions using Integrated Gradients (IG) and spatial uncertainty maps.

Accepted Sample: In correctly accepted cases, the model tends to focus on meaningful regions of the image, as shown by a concentrated IG heatmap. At the same time, the uncertainty map shows relatively low values, indicating that the model is confident in its decision.

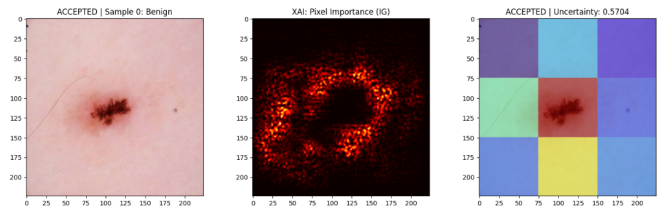

Fig. 1. Accepted Sample: A focused IG map and low uncertainty suggest a reliable prediction.

Rejected Sample: In contrast, rejected samples often display scattered or noisy attention in the IG visualization. The corresponding uncertainty map shows higher values, reflecting the model's lack of confidence. These cases are typically associated with noisy or ambiguous inputs.

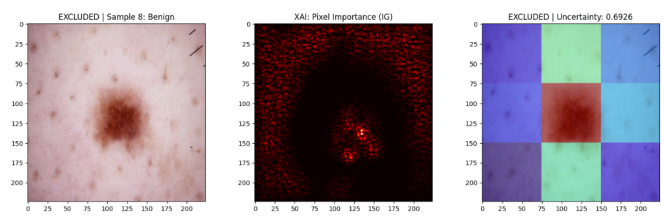

Fig. 2. Excluded Sample: Diffuse IG patterns and high uncertainty indicate an unreliable prediction.

B. Quantitative Results

The performance of the proposed approach is summarized below:

Metric	Value
Baseline Accuracy	80.30%
AUQantO Accuracy	82.45%
Threshold (λ)	0.685
Coverage	91.52%
Rejection Rate	8.48%
Accuracy Difference	0.3066

TABLE I
Performance Summary of Proposed Model

C. Observations

Several key patterns emerge from the results:

- Accepted predictions tend to be more accurate and confident.
- Rejected samples generally align with regions of high uncertainty.
- The model improves reliability by avoiding predictions when confidence is low.

Overall, the combination of uncertainty estimation and selective rejection leads to more dependable model performance.

VII. Analysis

The accepted samples show clear and focused attention around the lesion areas, suggesting that the model is picking up on clinically meaningful features. These samples also have lower uncertainty, indicating that the model is confident in its predictions. This consistency between where the model looks and how confident it is suggests that it is learning useful patterns from the data.

On the other hand, the rejected samples display scattered and unfocused attention in the Integrated Gradients maps. This indicates that the model struggles to identify strong distinguishing features in these cases. The higher uncertainty values further confirm this lack of confidence. Importantly, this is the desired behavior, as the model avoids making unreliable predictions on ambiguous inputs.

Overall, there is a clear relationship between uncertainty and prediction quality. Samples with sharp, meaningful feature attribution tend to have lower uncertainty, while unclear or noisy samples show both diffuse attention and higher uncertainty. This supports the effectiveness of the proposed framework in separating reliable predictions from unreliable ones.

VIII. Discussion

The proposed framework improves the trustworthiness of deep learning models by combining prediction, uncertainty estimation, and interpretability into a single system. By allowing the model to abstain from uncertain cases, it reduces the likelihood of incorrect predictions, which is especially important in medical applications.

Advantages:

- More reliable predictions through selective acceptance
- Better interpretability using Integrated Gradients and spatial uncertainty maps
- Ability to flag uncertain cases for expert review

Trade-offs:

- Some samples are rejected, which reduces overall coverage
- Higher computational cost due to multiple MC Dropout passes
- Added complexity from threshold optimization

IX. Limitations

Despite the encouraging results, there are a few limitations to consider:

- The relatively small dataset and batch size may limit how well the model generalizes
- The CNN architecture is simpler than many state-of-the-art alternatives
- MC Dropout increases inference time because it requires multiple forward passes
- The quality of explanations depends on how well the model has learned meaningful features

X. Conclusion

This work presents an uncertainty-aware framework for skin cancer classification using a two-stage CNN model. Monte Carlo Dropout is used to estimate prediction uncertainty, while COBYLA optimization determines an appropriate threshold for rejecting uncertain predictions.

The results show that selectively rejecting uncertain samples improves accuracy on accepted cases and reduces the risk of incorrect predictions. In addition, explainability methods like Integrated Gradients and spatial uncertainty maps provide useful insights into how the model makes decisions.

Overall, this approach improves both reliability and transparency, making it more suitable for real-world medical use. Future work could explore larger datasets, more advanced architectures, and faster uncertainty estimation methods.

Keywords

Uncertainty Estimation, MC Dropout, CNN, Explainable AI, Medical Imaging